MILITARY SPECIFICATION

THEOPHYLLINE, EPHEDRINE SULFATE, AND HYDROXYZINE HYDROCHLORIDE SYRUP, 1 pt (473 cc)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers Theophylline, Ephedrine Sulfate, and Hydroxyzine Hydrochloride Syrup, 1 pt (473 cc) (See 6.2). Potency: 60 Months.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-C-00186B	Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6505

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

AMERICAN PHARMACEUTICAL ASSOCIATION PUBLICATION

National Formulary.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE,
FOOD AND DRUG ADMINISTRATION

Federal Food, Drug, and Cosmetic Act and
Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

Pharmacopeia of the United States.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Shall be a syrup containing in each 5 cc, the following ingredients, within the designated assay limits:

Theophylline, Anhydrous	32.5 mg
Ephedrine Sulfate	6.25 mg
Hydroxyzine Hydrochloride	2.5 mg
Alcohol v/v	5.0%
Suitable vehicle, including coloring agent(s) (shall not contain FD & C Yellow - No. 5), flavor(s), and sweetening agent(s)	q.s.

In addition, suitable preservative(s), stabilizer(s), or other ingredient(s) may be added to insure stability of the product in compliance with the requirements stated herein.

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3.1.1 Description. The finished preparation shall be a pleasant-tasting solution.

3.1.2 Assay. The finished preparation shall assay to contain not less than 90.0 percent and not more than 110.0 percent of the required amounts of Theophylline, Ephedrine Sulfate, Hydroxyzine Hydrochloride, and Alcohol, when determined as specified in 4.4.

3.2 Active ingredients.

3.2.1 Theophylline, anhydrous. The anhydrous theophylline used in the preparation of the finished syrup shall be in accordance with the tests, standards, and requirements of the U.S.P., including any supplements or revisions thereto.

3.2.2 Ephedrine Sulfate. The ephedrine sulfate used in the preparation of the finished syrup shall be in accordance with the tests, standards, and requirements of the U.S.P., including any supplements or revisions thereto.

3.2.3 Hydroxyzine Hydrochloride. The hydroxyzine hydrochloride used in the preparation of the finished syrup shall be in accordance with the tests, standards, and requirements of the N.F., including any supplements or revisions thereto.

3.3 Other ingredients. All other ingredients used in the finished syrup shall be of U.S.P. or N.F. quality, or if not included in either of these compendia, they shall be suitable for use in the product.

3.4 Delivery. Not less than 56 months of the expiration dating period shall remain at time of delivery to the Government.

3.5 Workmanship. Workmanship shall be first class throughout. The material and its immediate containers shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 5 years after delivery of the supplies to which such records relate.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Lot. For purpose of this specification, a lot, batch, or control is that single, uniform, and homogeneous quantity of the finished syrup produced from one formulation, subjected to the same compounding and manufacturing operations, and filled into final containers.

4.3 Sampling. Sampling shall be conducted in accordance with the procedures set forth in Military Standard MIL-STD-105, with an acceptable quality level (AQL) of 1.0 percent defective for major defects and 2.5 percent defective for minor defects.

4.4 Tests.

4.4.1 Theophylline assay. The theophylline content in the syrup shall be determined as follows:

Standard preparation.

Quantitatively and stepwise, prepare a standard solution containing approximately 6.5 mcg of U.S.P. Theophylline Reference Standard in 0.IN Sodium Hydroxide.

Procedure.

Accurately transfer 5 ml of the finished syrup into a 125 ml separatory funnel. Add 25 ml of 40 percent w/v ammonium sulfate solution and extract four times with 40 ml portions of chloroform. Backwash the chloroform with 10 ml of 40 percent ammonium sulfate solution and transfer the 10 ml to the aqueous layer. Discard the aqueous layer.

Transfer the chloroform layer to a 250 ml separatory funnel and backwash with 35 ml of 5N sodium hydroxide. Filter the chloroform portion through anhydrous sodium sulfate into a 250 ml volumetric flask. Dilute to volume with chloroform and retain for the Hydroxyzine Hydrochloride assay (see 4.4.3).

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Transfer the 5N sodium hydroxide layer into a 250 ml volumetric flask and dilute to volume with purified water and mix well. Pipet 5 ml of the sample into a 100 ml volumetric flask. Add 30 ml of purified water and dilute to volume with 0.1N sodium hydroxide and mix well.

Using 1 cm cells, concomitantly determine the absorbances of the standard and sample, with a suitable spectrophotometer, at the maxima, at about 320 mu and 272 mu, using 0.1N sodium hydroxide as a blank.

Calculate the percent Theophylline as follows:

Calculation:

$$\text{Percent of required amount} = \frac{Au}{As} \times C \times \frac{100}{5} \times \frac{250}{32.5} \times 100$$

Where Au = Absorbance of sample at 272 mu - absorbance at 320 mu
As = Absorbance of standard at 272 mu - absorbance at 320 mu
C = Concentration of standard in mg/ml.

4.4.2 Ephedrine sulfate assay.

Reagents.

Carbon disulfide-pyridine reagent. Accurately measure and mix 25 ml of spectrographic grade pyridine and 65 ml of Isopropyl Alcohol. Immediately prior to use, add 35 ml of carbon disulfide, A.R., and mix well.

Cupric chloride reagent. Dissolve 60 mg of cupric chloride dihydrate in 125 ml of purified water in 250 ml volumetric flask. Dilute to volume with spectrographic grade pyridine.

Standard solution. Accurately weigh approximately 70 mg of ephedrine sulfate standard. Transfer to a 100 ml volumetric flask, dissolve and dilute with purified water and mix well. Transfer 10 ml into a 25 ml volumetric flask and dilute to volume with 0.4 percent sodium bicarbonate solution. (Prepare fresh daily).

Sample Solution.

Accurately weigh an accurately measured 20 ml aliquot of the finished syrup and transfer to a 100 ml volumetric flask. Dilute to volume with 0.4 percent sodium bicarbonate solution and mix well.

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Procedure.

To separate 40 to 50 ml glass-stoppered centrifuge tubes, add 3 ml of the standard solution, 3 ml of sample solution, and 3 ml of 0.4 percent sodium bicarbonate (blank). To each tube, add 10 ml of carbon disulfide-pyridine reagent. Start timing the reaction. Then in succession add 5 ml of cupric chloride reagent and 6 ml of 10 percent acetic acid solution, taking care to stopper, and swirl gently after each addition. Next add 10 ml of benzene, stopper and agitate the solution by inverting it four times. Centrifuge the tubes at 2000 r.p.m. for three minutes. Carefully pipet 10 ml of the upper phase into a 25 ml volumetric flask and dilute to volume with isopropyl alcohol. After a lapse of exactly 35 minutes, measure the absorbances of the standard and sample solutions, with a suitable spectrophotometer, at approximately 440 mu, using the blank solution to set the instrument.

Calculate the percent of the required amount of ephedrine sulfate, as follows:

Percent of required amount =

$$\frac{Au}{As} \times \frac{\text{Std Wt (mg)}}{\text{Sample Wt.(gram)}} \times \frac{0.4 \times \text{Sp. G.} \times 5}{6.25} \times 100$$

Where:

Au = Absorbance of sample at 440 mu
As = Absorbance of standard at 440 mu
Std Wt (mg) = Weight of standard in mg
Sample Wt (gram)= Weight of sample in grams
Sp. G. = Specific gravity of the syrup.

4.4.3 Hydroxyzine hydrochloride assay.

Standard solution.

Quantitatively and stepwise prepare a standard solution containing approximately 10 mcg of Hydroxyzine Hydrochloride Reference Standard per ml in 0.1N hydrochloric acid.

Procedure.

Pipet 20 ml of the chloroform solution retained from the Theophylline assay into a 100 ml beaker and evaporate to dryness on a steam bath. Allow to cool. Quantitatively transfer the residue to a 20 ml volumetric flask with aid of 0.1N hydrochloric acid and dilute to volume with 0.1N hydrochloric acid.

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Using 1 cm cells, concomitantly determine the absorbance of the standard and sample with a suitable spectrophotometer, at the maxima of about 232 mu and 250 mu, using 0.1N hydrochloric acid as the blank.

Calculate the percent Hydroxyzine Hydrochloride as follows:

$$\text{Percent of required amount} = \frac{Au}{As} \times \frac{C \times 250}{2.5} \times 100$$

Where Au = Absorbance of sample at 232 mu - absorbance at 250 mu
As = Absorbance of standard at 232 mu - absorbance at 250 mu
C = Concentration of standard in mg/ml.

4.4.4 Alcohol assay. Proceed as directed in the U.S.P. Gas-liquid Chromatographic method for the determination of Alcohol.

4.5 Examination of preparation for delivery. An examination shall be made to determine whether the packaging, packing and marking comply with the requirements of section 5. Defects shall be scored as specified but not limited to those in table I. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of defects per hundred units.

Table I. Examination of preparation for delivery

Examine	Defect
Containers	Not as specified.
Contents	Not as specified.
Markings	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Component missing or damaged.
Workmanship	Bulging or distortion of containers. Cushioning inadequate, improper or missing.

4.5.1 Examination of closure and reinforcement for containers. When shipping containers are required to comply with PPP-B-636, examination for defects in closure and reinforcement shall be in accordance with the appendix to that specification.

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5. PREPARATION FOR DELIVERY

5.1 Shall be in accordance with all applicable requirements of Interim Federal Specification PPP-C-00186B, dated 16 June 1972, and as specified herein.

5.1.1 Immediate containers. Shall comply with the following: Shall be a commercially available immediate container considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

5.1.2 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act, and shall include the information required below:

5.1.2.1 Immediate container. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.

(b) the expiration date.

5.1.2.2 Unit packages. Each unit package label shall bear the following:

(a) labeling information in accordance with commercial practice.

(b) the National Stock Number.

NOTE: The National Stock Number shall not be preceded by any prefix.

Strip labeling may be used.

Label adhesion test shall not apply.

If a unit package is not supplied, the National Stock Number shall appear on the immediate container label.

5.2 Packaging and Packing.

5.2.1 Unit of issue. One bottle containing one pint, as specified, constitutes one unit of issue.

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5.2.2 Unit package. At the option of the contractor, a unit package may be furnished. If furnished each unit shall be packaged as specified in 5.2.5 of PPP-C-00186B.

5.2.3 Procedure code. Procedure code No. 7 as specified in Table I of PPP-C-00186B shall apply.

5.3 Marking.

5.3.1 Intermediate package. Each intermediate package shall be marked in accordance with 5.5.3 and 5.5.8.1 of PPP-C-00186B. Marking shall include the legend:

"STORE AT CONTROLLED ROOM TEMPERATURE (59°-86°F)".

5.3.2 Exterior container. Exterior container shall be marked in accordance with 5.5.4 and 5.5.8.1 of PPP-C-00186B. Marking shall include the legend:

"STORE AT CONTROLLED ROOM TEMPERATURE (59°-86°F)".

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) the National Stock Number.
(c) Selection of applicable levels of packaging and packing shall be as specified in the contract or order.

6.2 This specification covers the following item:

National Stock No.	Item Identification
6505-00-134-2007	THEOPHYLLINE, EPHEDRINE SULFATE, AND HYDROXYZINE HYDROCHLORIDE SYRUP, 1 pt (473 cc)

6.3 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet Military requirements.

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Custodians:
Army - MD
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Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6505-1344

Review information is current as of date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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MIL-T-37167

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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